1 Intent plane: one typed object per sentence, compiled deterministically

Natural-language intents

- finance may reach database over PostgreSQL (TCP 5432).
- dev-lab may use any TCP port towards ops, except Telnet (TCP 23), which must be blocked.
- cameras is isolated: it may only reach nvr on RTSP (TCP 554), and nothing else.
- guest-wifi must never reach finance, regardless of any other rule.

zone inventory: zone → router, prefixes
(Topology Zoo graph, ClassBench prefixes)

type-selection rules + 7 placeholder examples; one call per sentence

Qwen3-8B (local, llama.cpp, greedy, JSON)

invalid → re-ask

typed IR (one per sentence)

- allow(*s, d, S*)
- allow_except(*s, d, b, E*)
- block(*s, d*)
- isolate(*s, A*)
- public(*d, S*)

Deterministic compiler
precedence:
block ≻ except ≻ allow ≻ isolate ≻ public

zone policy Π (first match)

`guest → fin *`	D
`dev → ops tcp/23`	D
`fin → db tcp/5432`	P
`dev → ops tcp/*`	P
`cam → nvr tcp/554`	P
`cam → * *`	D

implicit default deny between zones

deterministic validator (no second LLM opinion): every object is checked before compilation

- type ∈ 5 types
- zones ∈ inventory
- services ∈ catalogue
- fields complete

on failure: error message back to the model, re-ask ≤ 2

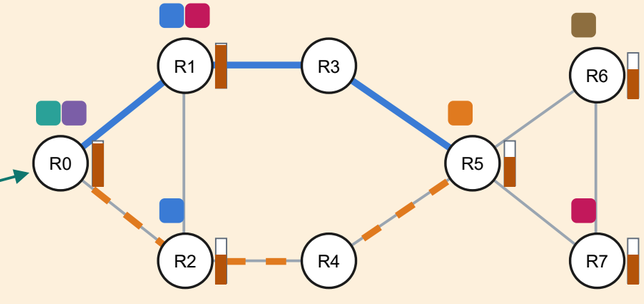
2 Disjoint rule blocks

Π restricted to one (source prefix, destination prefix) pair + explicit trailing deny

- b_1 10.0.16.0/20 → 10.0.66.0/25 : 3 rules
- b_2 10.0.16.0/20 → 10.0.32.0/21 : 1 rule (deny)
- b_3 10.0.4.0/22 → 10.0.66.0/25 : 2 rules
- b_4 10.0.8.0/22 → 10.0.48.0/20 : 4 rules
- b_5 10.0.64.0/26 → 10.0.4.0/22 : 1 rule
- b_6 10.0.72.0/24 → 10.0.8.0/22 : 2 rules

blocks never overlap in header space and are idempotent ⇒ a block may sit on **any** router of the pair's paths, duplicates are harmless; cost c_b = #entries

3 Capacity-aware placement over all ECMP paths (ILP)



path π_1 --- path π_2 (equal cost) ▮ router load / C_r ■ placed block

$$\min \Sigma_{b,r}\, c_b\, x_{br}\,(1 + \alpha\, h_{br})$$

$$\Sigma_{r \in \pi}\, x_{br} \geq 1 \quad \forall\, \pi \in \mathrm{ECMP}(b)$$
$$\Sigma_b\, c_b\, x_{br} \leq C_r \quad \forall\, r$$
$$x_{br} \in \{0, 1\}$$

every equal-cost path crosses a holder; per-router entry budget; early drop (h = hop index)

witness → re-place

4 Render, verify, emulate

```
*filter
:FORWARD ACCEPT
-A FORWARD -m conntrack ...
-A FORWARD -s 10.0.16.0/20
  -d 10.0.66.0/25 -p tcp
  --dport 5432 -j ACCEPT
...
```

iptables-restore per router

Z3, full 5-tuple (src 32 b, dst 32 b, proto 8 b, port 16 b)
$\forall\, s, d, \forall\, \pi \in \mathrm{ECMP}(s, d)$:
$\wedge_{r \in \pi}\, \mathrm{dev}_r(h) \equiv \Pi(h)$
unsat ⇒ verified · sat ⇒ packet witness

GNS3 VM testbed
FRR 10.2 routers: OSPF, ECMP (L4 hash),
iptables; zone hosts as namespaces;
UDP, TCP-SYN, ICMP probes captured at the destination

KPIs: header-space equivalence, feasible capacity, entries / duplicates, drop position, solve and verification time, deployment latency, probe conformance and leaks